

Conceitos de Diversidade Alfa e Beta de Espécies - Similaridade Entre Habitats

Diversidade Alfa (local): combina riqueza e homogeneidade de abundâncias

Indices de Shannon, Simpson

Índice de Diversidade de Shannon:

The formula is derived from Claude Shannon's work on information entropy, measuring the uncertainty in predicting the species identity of a randomly chosen individual. Higher diversity means higher uncertainty. The maximum value of the Shannon diversity index is not fixed, but rather equals the natural log of the total number of species in the community (species richness). This maximum occurs only when all species have perfectly equal abundance. Typically, ranges between 1.5 and 3.5, though it can exceed 5 in highly diverse

Índice de Diversidade de Simpson:

Mathematically, it represents the probability that two individuals randomly selected from a sample will belong to the same species. The maximum value of the Simpson's Index of Diversity is approaching 1, representing infinite diversity or perfect evenness, while 0 represents no diversity. The value ranges from 0 to 1, with higher values indicating greater diversity. In practical terms, the closer the result is to 1, the more diverse the community is, with the highest possible diversity occurring when all species are equally represented.

Diversidade entre habitats (beta) % de similaridade, distância geométrica

Bray-Curtis, Jaccard, Sorensen, Morisita-Horn (bom estimador)

Diversidade Beta combina substituição de espécies e riqueza entre habitats

Turnover and nestedness are two additive components of species beta diversity that explain how community composition changes across space or time. Turnover (species replacement) measures the swapping of species between sites, often driven by environmental filtering, while nestedness (richness difference) occurs when species-poor sites are subsets of species-rich sites, often due to ordered extinction/colonization

Key Components and Synonyms:

Turnover: Often called species replacement or the Simpson dissimilarity index (Simpson 1943).

Nestedness: Also referred to as nestedness-resultant dissimilarity, describing richness-related differences or species loss/gain.

Total Beta Diversity: Defined by Sorensen dissimilarity, representing the sum of both components.

Usage Examples and Significance:

Conservation Planning: High turnover suggests that many different sites need protection to cover all species. High nestedness suggests targeting the most species-rich sites for conservation.

Macroecology: Turnover is often the dominant component of beta diversity, especially over large distances, while nestedness becomes relevant in isolated systems.

Community Assembly: Turnover is frequently associated with environmental sorting, while nestedness is linked to habitat degradation causing species loss.

Key Differences Summary:

Turnover (Replacement): Site A has species X; Site B has species Y.

Nestedness (Subset): Site A has species X, Y, Z; Site B has only species X.